or

$$\mathbf{H}|\psi(x)\rangle \qquad \Longrightarrow \qquad -\frac{\hbar^2}{2}\frac{\partial^2\psi(x)}{\partial x^2} + \frac{1}{2}\omega^2 x^2\psi(x). \quad (10.12)$$

We're using partial derivatives because in general ψ also depends on another variable, *time*. Time is not an operator and does not have the same status as x, but the state-vector does change with time, and we therefore treat time as a parameter. The partial derivative indicates that we're describing the system "at a fixed time."

10.3 THE SCHRÖDINGER EQUATION

Eq. 10.12 shows how the Hamiltonian operates on ψ. Now, let's put it to work. As we said in the previous section, one of its jobs is to tell you how the state-vector changes with time. So let's write out the time-dependent Schrödinger equation:

$$i\frac{\partial\psi}{\partial t} = \frac{1}{\hbar}\mathbf{H}\psi.$$

Substituting for **H** using 10.12, we get

$$i\frac{\partial\psi}{\partial t} = -\frac{\hbar}{2}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2\hbar}\omega^2 x^2\psi. \qquad (10.13)$$

This equation says that if you know ψ (both the real and imaginary parts) at some particular time, you can predict what it will be at a future time. Notice that the equation is complex—it contains i as a factor. This means that even if ψ starts out being real-valued at time $t = 0$, it will very shortly develop an imaginary part. Any solution ψ must therefore be a complex function of x and t.